

Date 2080/12/13
Class BSc 2nd Year
Roll No 68
Shift Day

Physics Practical Sheet
Prithvi Narayan Campus

Experiment No 03
Group
Sub Physics
Set

Object of the Experiment (Block Letters) : TO CONSTRUCT NOR AND NAND GATE
AND HENCE VERIFY ITS TRUTH TABLE BY DTL.

APPARATUS REQUIRED:

- | | | |
|------------------|-----------------|-----------------|
| (i) Diode | (ii) Transistor | (iii) Lead |
| (iv) Bread board | (v) Wires | (vi) Multimeter |

THEORY:

① NOR GATE:

In fact; it is a NOT-OR gate. It can be made out of an OR gate by connecting an inverter in its output shown in the figure.. The output operation is given by ;

$$Y = \overline{A+B}$$

Logic Operation

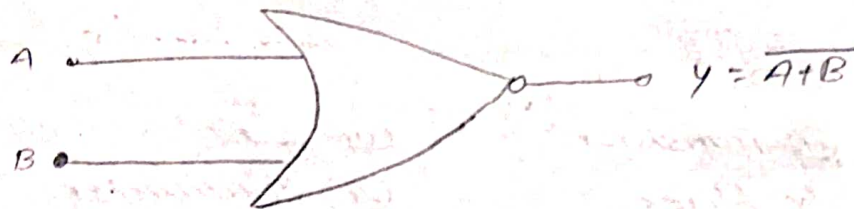
An NOR gate will have an output of 1 only when all its input are 0. Obviously, if any input is 1, the output will be zero. Alternatively, in an NOR gate, output is true only when all inputs are false.

Truth Table

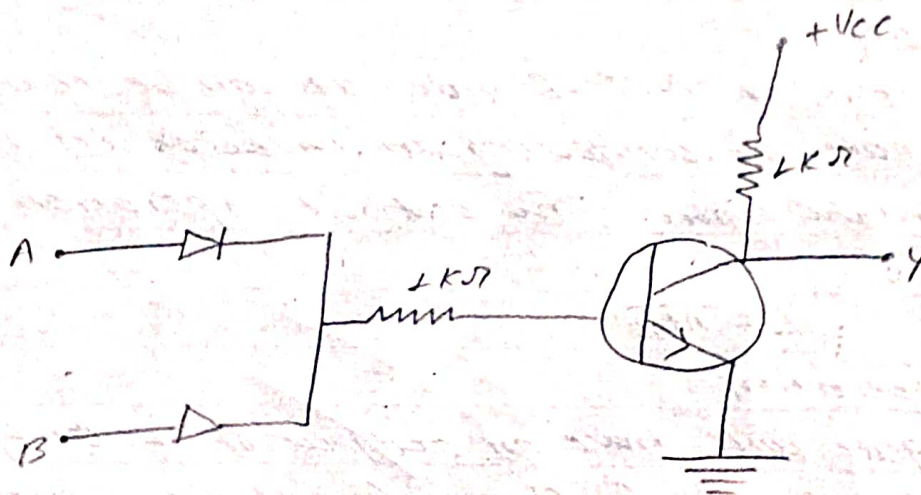
A	B	$Y = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

The truth table for - 2-input NOR gate is shown in the Table. It is observed that output 'Y' is just a reverse OR that is shown in the table.

① NOR GATE



Symbol of NOR gate



Electrical circuit of NOR gate

① NAND GATE:

It is in fact; an NOT-AND gate. It can be obtained by connecting a NOT gate in the output of an AND gate. Its output is given by the Boolean equation

$$Y = \overline{A \cdot B}$$

This gate gives an output of 1, if its input are not 1. In other words; it gives an output 1 if either A or B or both are 0.

The truth table for -2-input NAND gate is given below. It is just opposite of the truth table of AND gate. It is so because NAND gates perform reverse function of an AND gate.

Truth Table

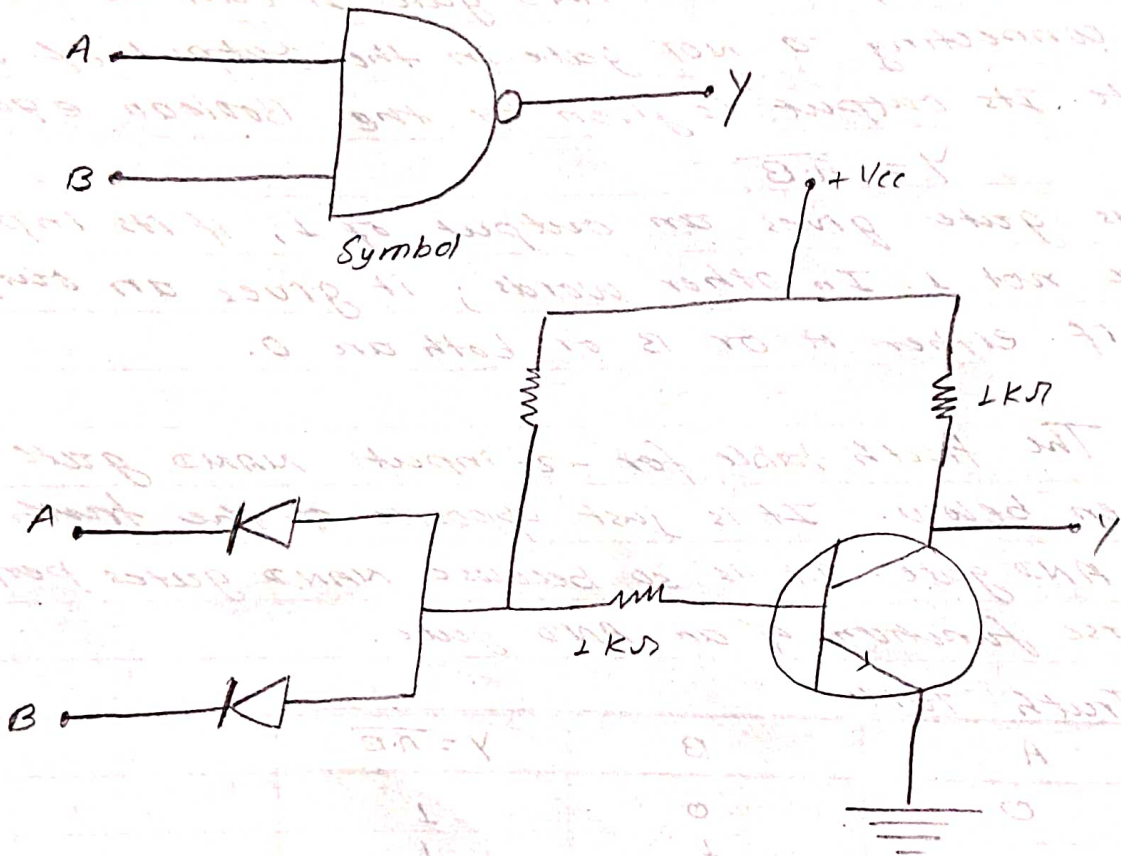
A	B	$Y = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

OBSERVATION:

② NOR Gate

S.N.	Input		Output
	A	B	$Y = \overline{A+B}$
1	0	0	1
2	0	1	0
3	1	0	0
4	1	1	0

② NAND GATE



Electrical circuit of NAND Gate

Input	Output
A	B
0	0
0	1
1	0
1	1

$Y = A \cdot B$

(II) NAND Gate:

S.N	Input		Output
	A	B	$Y = \overline{A \cdot B}$
1	0	0	4.10
2	0	4.20	3.74
3	4.20	0	3.66
4	4.20	4.20	0.02

RESULT:

From the above experiment, the functions of simple logic gates are observed.

PRECAUTIONS:

- (i) Common emitter base should be connected properly.
- (ii) Connection of diode should be right.
- (iii) Millimeter reading/voltage current should be noted as an end result input.